

Some Thoughts on Research

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an academic...



teaching



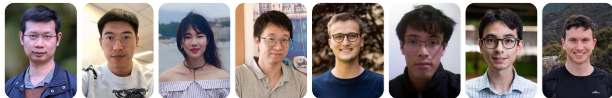
proving



hacking



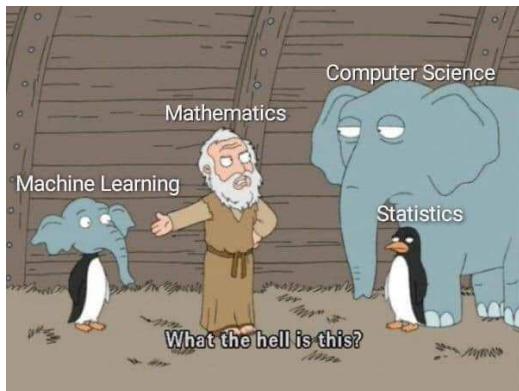
babysitting



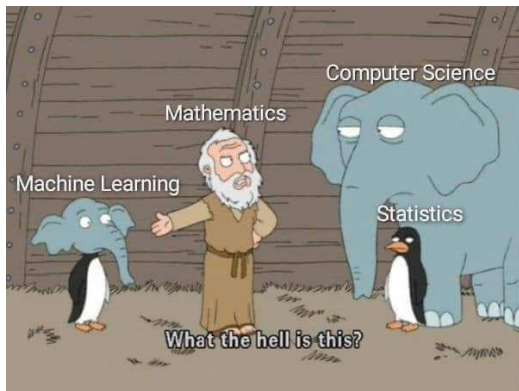
supervising some very intelligent people

cartoons generated by AI

big picture #1



big picture #1



Looks like the person who made this wanted to say machine learning is just taking something from computer science and something from statistics, and ended up being not recognised by either community and scorned by mathematics. . .

This might be the view held by quite a few people in the past, but perhaps not any more - the picture would be something like machine learning is a rocket, pulling mathematics, computer science, statistics, biology, physics, ..., up to the sky :D

email to a colleague

big picture #2



an attempt by AI to draw my picture

big picture #3 (my work)



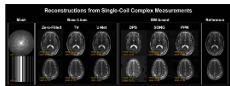
autonomous driving



routing behavior analysis

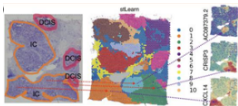


fishery stock assessment

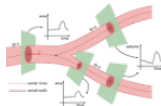


MRI reconstruction

theoretically grounded practical
algorithms
for
learning & decision-making

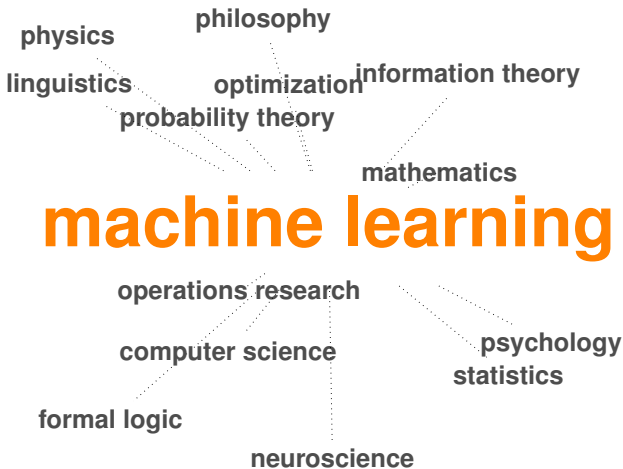


spatial transcriptomics

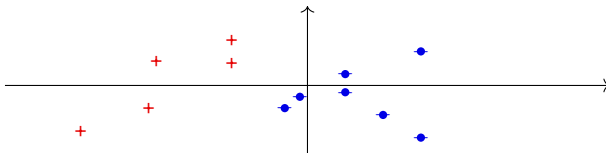


haemodynamics modeling

big picture #4 (synergy)

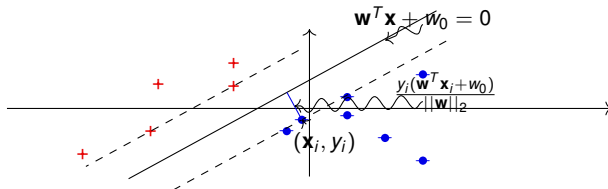


an example



question. what's the best line to separate the red and blue points?

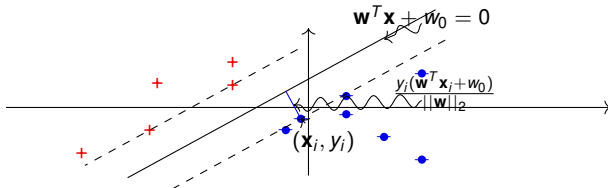
an example



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intuition. support vector machines: the one farthest away from all points.

an example



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intuition. support vector machines: the one farthest away from all points.

math.

$$\begin{aligned} \min_{\mathbf{w}, w_0} \quad & \frac{1}{2} \|\mathbf{w}\|_2^2 \\ \text{s.t.} \quad & y_i(\mathbf{w}^T \mathbf{x}_i + w_0) \geq 1, \quad i = 1, \dots, n. \end{aligned}$$

algorithm. how to compute a solution to the optimization problem? $\Rightarrow$ SMO and more...

extensions. what happens if there is no separating line? $\Rightarrow$ slack variables, kernel trick

theory. does this work? $\Rightarrow$ statistical learning theory

practice. applications on real classification datasets

some thoughts

#1 find good questions

they are important

you want to solve them

neither too difficult nor too easy

...

In mathematics the art of proposing a question must be held of higher value than solving it. Georg Cantor

#2 a failed attempt is a useful finding

a failed attempt eliminates a possible solution

analyzing how things fail can lead to insights on how to make things work

Once you eliminate the impossible, whatever remains, no matter how improbable, must be the truth. Arthur Conan Doyle

#3 balance intuition and rigor

intuition allows us to quickly see a path to a solution, but it may be wrong

rigor is needed for correctness, but often tedious and hides the idea

It is by logic that we prove, but by intuition that we discover. Henri Poincaré

machine learning research: big pictures, an example

some thoughts

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